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# Transplant benefit-based offering of deceased donor livers in the United Kingdom

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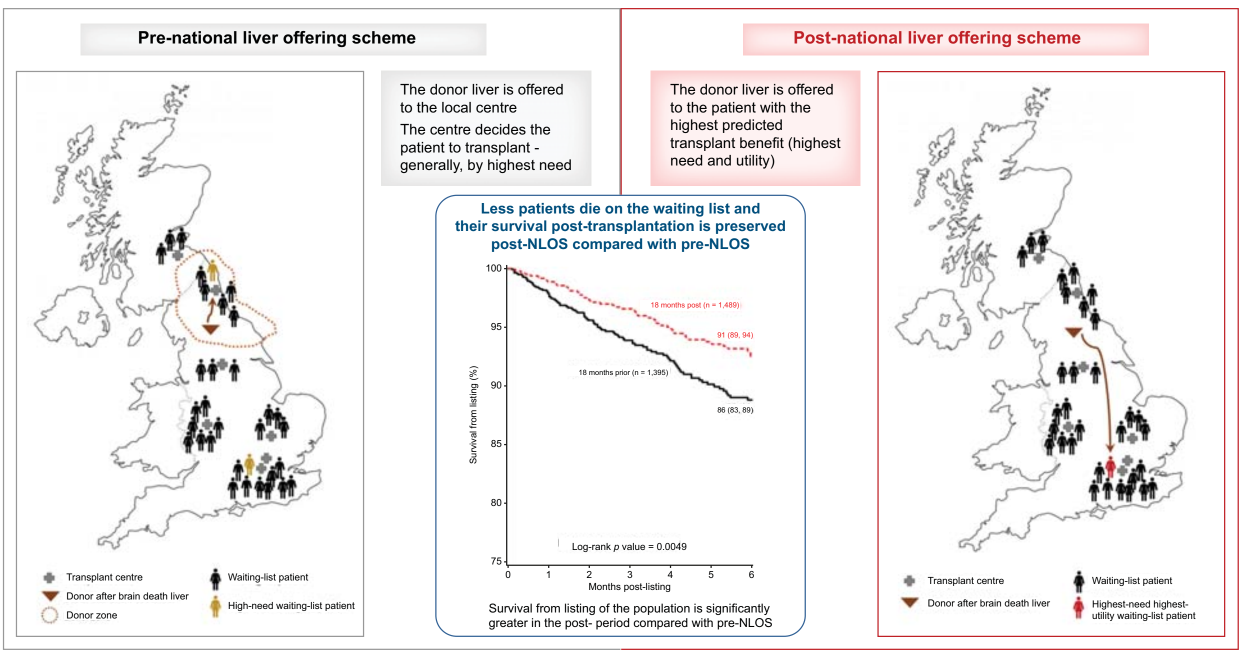
## Graphical abstract




## Highlights

- In 2018 the UK introduced a benefit-based scheme to offer deceased donor livers.
- Benefit is predicted using a patient's need for, and the utility of, a transplant.
- The novelty of the scheme relates to the incorporation of predicted utility.
- The scheme reduces waiting-list mortality and preserves post-transplant outcomes.
- Life-year gain from transplantation is maximised for the waiting list population.

## Impact and implications

The National Liver Offering Scheme (NLOS) was introduced in the UK in 2018 to increase transparency of the deceased donor liver offering process, maximise the overall survival of the waiting list population, and improve equity of access to liver transplantation. To our knowledge, it is the first scheme that offers organs based on statistical prediction of transplant benefit: the transplant benefit score. The results are important to the transplant community – from healthcare practitioners to patients – and demonstrate that, in the first two years of NLOS offering, waiting list mortality fell while post-transplant survival was not negatively impacted, thus delivering on the scheme's objectives. The scheme continues to be monitored to ensure that the transplant benefit score remains up-to-date and that signals that suggest the possible disadvantage of some patients are investigated.

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# Transplant benefit-based offering of deceased donor livers in the United Kingdom

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**Background & Aims:** The National Liver Offering Scheme (NLOS) was introduced in the UK in 2018 to offer livers from deceased donors to patients on the national waiting list based, for most patients, on calculated transplant benefit. Before NLOS, livers were offered to transplant centres by geographic donor zones and, within centres, by estimated recipient need for a transplant.
**Methods:** UK Transplant Registry data on patient registrations and transplants were analysed to build statistical models for survival on the list (M1) and survival post-transplantation (M2). A separate cohort of registrations – not seen by the models before – was analysed to simulate what liver allocation would have been under M1, M2 and a transplant benefit score (TBS) model (combining both M1 and M2), and to compare these allocations to what had been recorded in the UK Transplant Registry. The number of deaths on the waiting list and patient life years were used to compare the different simulation scenarios and to select the optimal allocation model. Registry data were monitored, pre- and post-NLOS, to understand the performance of the scheme.
**Results:** The TBS was identified as the optimal model to offer donation after brain death (DBD) livers to adult and large paediatric elective recipients. In the first 2 years of NLOS, 68% of DBD livers were offered using the TBS to this type of recipient. Monitoring data indicate that mortality on the waiting list post-NLOS significantly decreased compared with pre-NLOS ($p$ <0.0001), and that patient survival post-listing was significantly greater post- compared to pre-NLOS ($p$ = 0.005).
**Conclusions:** In the first two years of NLOS offering, waiting list mortality fell while post-transplant survival was not negatively impacted, delivering on the scheme's objectives.



## Introduction

Liver transplantation is currently the only treatment for end-stage liver failure. The number of deceased liver donors and transplants increased in the UK over recent years (except during the COVID-19 pandemic) with 823 transplants performed in 2021/22 compared to 784 in 2012/13. Despite this, an important proportion of patients die or become too sick while waiting. Of 1,046 new elective liver-only registrations in the UK between April 2019 and March 2020, 52 (5%) patients died within 6 months after joining the transplant list,[2] with equivalent figures being 82 (8%) and 71 (7%) in 2016/17 and 2017/18, respectively.[3,4]

The National Liver Offering Scheme (NLOS) was introduced in the UK on 20 March 2018 for all donation after brain death (DBD) livers, with the aim of nationalising liver offering across the UK, increasing transparency and objectivity of the offering process, maximising the overall survival of the waiting list population (whether before or after transplant), and improving equity of access to transplantation. Before NLOS, livers for adult elective recipients (who comprise most recipients waiting for a transplant) were distributed based on locality of the donor and past transplant activity of centres. Donor hospitals belonged to a transplant centre's locality (or *donor zone*) based on the share of patients registered for a transplant in that centre relative to the national number of elective adult registrations; share of donors and share of registrations was balanced. Upon receiving an offer, the centres used to prioritise recipients within their local waiting list using the United Kingdom End-Stage Liver Disease (UKELD) score[5] (supplementary data), an indicator of disease severity where higher values correspond to patients who are more seriously ill (see Table S1 for a comparison of the UKELD score with other commonly used scores). Offers were made to centres rather than to named individuals. If the local centre declined the offer for all patients on its list, the organ would be offered to centres in other regions in the country, based on proximity (for donation after circulatory death [DCD]) or lowest recent transplant activity (for DBD; Fig. S1). It was recognised that recipients with similar poor predicted survival on the waiting list and a good prognosis after receiving the transplant (utility) did not have equity of access to

Keywords: Liver transplantation; transplant benefit; equity of access; organ donation after brain death; organ donation after circulatory death; survival on the waiting list; survival post-transplantation; organ offering and allocation.
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transplantation due to the geographic nature of the old offering scheme.

Early proposals to develop a 'Universal Liver Transplant Allocation Scheme' were being discussed by the UK liver transplant community as early as 2008.[6] A first version of the NLOS was presented to the community in 2012 at a Consensus Conference where it was concluded that more statistical work was needed before any change could be introduced and that the concept of 'utility' should be investigated. A working unit of the UK Liver Advisory Group (LAG) was formed in 2013 to compare the existing centre-based scheme with prioritisation of recipients by risk of death or drop out from the waiting list due to clinical deterioration (need-based offering), prioritisation by probability of survival after transplantation (utility-based offering) and prioritisation by net life years gained from transplantation (benefit-based offering). The journey to developing, implementing, and monitoring a NLOS for the UK has been lengthy and challenging. This paper describes the algorithms that were set up in the UK to offer DBD livers under the NLOS and our monitoring of them. It also briefly presents monitoring data on the offering of DCD livers, which have thus far continued to be offered outside the NLOS. We report NLOS performance up to the start of the COVID-19 pandemic, in March 2020, as liver donor and transplant activity in the UK was significantly impacted by the pandemic. Up-to-date NLOS performance data are available online.[1] We finish off by commenting on recent (post-pandemic) performance and describing recent developments introduced to the scheme in October 2022.

## Materials and methods

### Study populations

*NLOS development*

Two types of statistical models were developed; (1) survival on the waiting list – prioritising those with the shortest predicted survival and (2) survival post-liver transplantation – prioritising those with the longest predicted survival. The cohort for the *survival on the waiting list model* (M1) comprised all adult (aged ≥17 years) elective NHS Group 1[7] registrations for a liver-only transplant in the UK. The cohort for the *survival post-transplantation model* (M2) included all adult elective NHS Group 1 liver-only transplants in the UK using deceased donors (either DBD or DCD). The M1 and M2 cohorts were split into two sub-cohorts based on whether the recipient had cancer[1] as an indication for transplantation, or not, as reported at time of registration or transplant, respectively. For the development cohorts, the period of elective patient registrations was 1 January 2006 to 31 December 2012 for the non-cancer (NC) sub-cohort and 1 January 2009 to 31 December 2012 for the cancer (C) sub-cohort (as some cancer parameters were not collected previously). Registrations ending in living, or domino, donor transplantation as well as multi-organ registrations were excluded from M1. Data were obtained from the UK Transplant Registry (UKTR) held by NHS Blood and Transplant (NHSBT) as of 9 June 2014 for M1, and as of 6 July 2014 for M2 (cohort sizes: $n_{M1,NC} = 3,859$, $n_{M1,C} = 660$, $n_{M2,NC} = 2,495$, $n_{M2,C} = 430$). How NHSBT uses patient information is described in[22] and the supplementary methods.

A simulation cohort mimicked the allocation of liver donors to recipients according to hypothetical offering scenarios and compared these modelled approaches with the actual organ allocation observed for the organs utilised during the study period (DBD and DCD). The cohort comprised all UK adult elective NHS Group 1 registrations for a liver-only transplant active on the waiting list between 1 January and 31 December 2013, including new registrations made during 2013. A total of 51 registrations were excluded (see the supplementary data) resulting in a final simulation cohort of 1,287 registrations and 629 donors.

*Monitoring the impact of the NLOS*

Data on waiting list and transplants were obtained from the UKTR as at 20 March 2020 or 16 April 2020, depending on the feature of the offering scheme examined, with little expected variation in data configuration between these dates (Table 1). The *monitoring period* included 24 months prior to NLOS introduction (20 March 2016 to 19 March 2018) and 24 months post-NLOS (20 March 2018 to 19 March 2020), unless otherwise specified.

### Statistical methods

Further information on the software and methods used are provided in the supplementary CTAT table.

### NLOS development

*Modelling survival on the waiting list*

Cox proportional hazard regression methods were used to model time from patient registration to death on the list or removal from the list due to deteriorating condition, up to 5 years post-registration (M1), independently for the cancer and non-cancer sub-cohorts. Removals from the waiting list due to improved condition or registrations that had not yet ended at the time of analysis were censored. Patients who survived more than 5 years on the list were censored at 5 years as this length of follow-up was deemed appropriate for the purposes of the analysis. Registrations ending in transplantation were informatively censored because donor livers were allocated based on greatest need during the considered period. Therefore, transplanted patients are censored from the study when their risk of death is high which means that treating their time to transplant simply as non-events may result in overoptimistic survival estimates. Inverse probability of censoring weights were used to account for informative censoring,[8] as recommended by.[9,23] The probability of censoring was estimated from a survival model with a Weibull parameterisation where informative censoring (*i.e.* transplantation within 5 years) was the event. The explanatory variables in the Weibull model were identical to those used in M1. The inverse probability of censoring weights were used to weigh the contribution of each patient to the partial likelihood of M1 at each event time. To account for the additional uncertainty in the model specification, a robust "sandwich" estimate of the covariance matrix was introduced.

[1] Cancer indications for elective liver registration at the time of NLOS development were hepatocellular carcinoma (non-cirrhotic), hepatocellular carcinoma (cirrhotic), cholangiocarcinoma, hepatoblastoma, secondary hepatic malignancy and other primary hepatic malignancy.

**Table 1. Cohort selection and size used to monitor the impact of the NLOS.[†]**

| Monitoring feature | Data inclusion criteria | Size of cohort, n |
|---|---|---|
| Death on the transplant list and survival from listing | Adult elective NHS Group 1 liver patients active or suspended on the liver transplant list during a *reduced*[*] monitoring period of 18 months were included. | 2,875 |
| Registration activity | New NHS Group 1 registrations on the UK liver transplant list were included. | 4,725 |
| Offering | UK deceased donors (either DBD or DCD) whose liver was offered for transplantation, including offers to Dublin for super-urgent (but excluding elective) patients, during the monitoring period were included. Offers declined due to the patient accepting a previously offered liver were excluded. | 7,736 |
| Transplant activity | All UK liver transplants from deceased donors (either DBD or DCD) during the monitoring period were included. | 3,903 |

DBD, donation after brain death; DCD, donation after circulatory death; NLOS, National Liver Offering Scheme.
[†]The *monitoring period* included 24 months prior to NLOS introduction (20 March 2016 to 19 March 2018) and 24 months post-NLOS (20 March 2018 to 19 March 2020). Unless otherwise stated, registrations, offers, or transplants performed at Dublin, or for intestinal recipients who did not require a liver, or for NHS Group 2[7] recipients were excluded.
*The pre-NLOS period was *reduced* to between 20 March 2016 and 19 September 2017 and post-NLOS to between 20 March 2018 and 19 September 2019 (18 months instead of 24) to allow for 6-month follow-up.

An initial set of candidate explanatory variables was identified using clinical expertise. Variable selection from the candidate set was performed using a combination of stepwise selection[9] at the 5% significance level and clinical judgement. Variables with a skewed distribution were transformed using the natural logarithm, as appropriate. Renal support and patient location were missing in around 20% of cases. Multiple imputation (MI) with chained equations was used to impute these missing data.[10] The form of the imputation model was like the M1 model and included the missing variable not being imputed (*i.e.* renal support or patient location) as well as the survival time and censoring indicator. Twenty-one imputations were run with 50 burn-in iterations before each final imputation. The modal value out of the 21 imputed values for each patient with missing renal support or patient location was taken as the imputed datum. A sensitivity analysis that compares our modal value approach with an approach that analyses each imputed dataset separately and then combines parameter estimates by Rubin's rules[10] is shown in the supplementary data. We note that[24] presents an alternative imputation model for survival data (with non-informative censoring) which we have not implemented in our study as, according to performance metrics in,[24] the imputation model recommended by[24] and our imputation model do not perform dissimilarly. Once MI was completed, 6% (308/4827) of registrations were excluded due to other missing data to create the final cohorts $n_{M1,NC}$ and $n_{M1,C}$.

*Modelling survival post-transplantation*
Cox proportional hazard regression was used to model time from patient transplant to the earlier of patient death or graft failure up to 5 years post-transplant (M2), independently for the cancer and non-cancer sub-cohorts. Deaths with a functioning graft were included as events, whilst patients who were alive with a functioning graft were censored at the earliest of last follow-up or 5 years. A 5-year follow-up period post-transplantation was considered clinically appropriate for the purposes of the analysis.

An initial set of candidate explanatory variables was identified using clinical expertise. As before, variable selection was performed using a combination of stepwise selection[9] and clinical judgement. Skewed variables were transformed using the natural logarithm. MI with chained equations was used to impute values missing for diabetes in 26% of recipients. A binary logistic regression model was developed to identify the variables for the imputation model, using the recipient variables in M2 as well as recipient blood group, ethnic group and BMI. Variable selection was done using stepwise selection at the 10% significance level. The variables included in the imputation model were recipient diabetes, survival time post-transplant, censoring indicator, disease group, age, serum bilirubin[2], potassium, presence of ascites, BMI, presence of hepatitis C and ethnic group. Five interaction terms were also included: age*serum bilirubin,[2] age*presence of ascites, serum bilirubin[2]*BMI, disease group*serum bilirubin[2] and disease group*potassium. An alternative imputation model could follow[24] but we chose not to do this for similar reasons to M1. After MI, 16% (559/3,484) of observations were excluded due to other missing data to create the final cohorts $n_{M2,NC}$ and $n_{M2,C}$.

The predictive ability of both types of models (M1 and M2) was assessed using the Gönen and Heller's Concordance statistic (C-statistic), which ranges between 0.5 (no predictive ability) and 1.0 (perfect predictive ability).[11] The C-statistic standard error was used to calculate a 95% CI for the C-statistic.

*The transplant benefit score*
Let $i$ denote the $i$ th patient[3] awaiting a liver transplant on the elective waiting list and suppose that there are $n$ such patients, so that $i = 1, ..., n$. The transplant benefit score (TBS) for recipient $i$ is the difference between their expected survival following transplantation with the graft on offer and without a transplant:

$$TBS_i = M2_i - M1_i \quad (1)$$

where $M1_i$ is the patient's expected survival without a transplant (survival on the list) and $M2_i$ is the patient's expected survival with a

[2] Log-transformed.

[3] *Patient* is defined as any *eligible adult*, *small adult* or *large paediatric* on the elective liver-only or combined liver/kidney transplant waiting list and with an indication for transplantation in the chronic liver disease (CLD), hepatocellular carcinoma (HCC) or variant syndrome in the context of CLD (diuretic resistant ascites and/or chronic hepatic encephalopathy) groups. 'Eligible' is defined as any NHS group 1 patient of the type above within a ±20 kg weight range relative to the donor's weight and blood group compatible with the donor. An 'adult' liver recipient is a patient aged 17 years or over at the time of registration and not a small adult. A 'small adult' liver recipient is a patient aged 17 years or over at the time of registration with a body weight of 40 kg or less and dual-listing option specified at the time of offer. A 'large paediatric' liver recipient is a patient aged 16 years or less at the time of registration, with a body weight of 40 kg or more and dual-listing option specified at the time of offering.[7,12]

transplant (survival post-transplantation) over a 5-year period. From equation (1), a patient with low expected survival on the list (high need) and large expected survival post-transplantation (high utility) produces a high TBS, whereas a patient with high expected survival on the list and low expected survival post-transplantation achieves a low TBS. We use TBS interchangeably with M3. For each patient *i* on the waiting list who is eligible for a liver on offer, two quantities are computed at time of organ offering: $M1_i$ and $M2_i$, as described in the supplementary data.

### Allocating livers under different simulated scenarios to find the optimal offering model

Computer simulations were run to mimic the allocation of liver donors to recipients under three different scenarios; based on need (the liver is allocated to the registrant with the shortest post-registration survival; M1), utility (the liver is allocated to the registrant with the longest post-transplant survival; M2), and benefit (the liver is allocated to the registrant with the highest TBS). These approaches were then compared with the actual allocation during 2013 (the period of the simulation cohort). It was assumed that the first recipient to be offered the liver in our simulations would accept it (hence our use of the word *allocation* during the NLOS development phase and *offering* after implementation).

Requirement to split the donor liver, donor-recipient blood group and weight compatibility (recipient weight ±20 kg of donor weight), as per agreed NHSBT policies at the time, were incorporated into the simulation. Both DBD and DCD donor livers were included. Patients who were transplanted in real-life but not transplanted under a scenario presented the problem of not having an observed death date on the list. We estimated a death date for these patients using M1 from the date on which the real-life transplant was performed. These patients were kept in the pool of potential recipients in the simulations until either an organ was allocated to them, or their predicted date of death was reached. Once a registrant was allocated an organ under a given model, they were excluded from the pool of potential recipients.

The primary outcomes to compare scenarios were number of deaths while on the transplant list and total population life years (in the pre- and post-transplant populations combined). Deaths on the list were counted as patients who died or were removed due to clinical deterioration, including the number of patients who died or were removed in real-life during 2013 without being allocated a liver in the simulations. It also included patients who were transplanted in real-life but not under a model and whose estimated death date on the list using M1 was during 2013. Population life years were calculated for all registrations in the cohort and depended upon the registration outcome. Life years for patients allocated a liver was their observed survival time from registration to date of notification of a liver offer plus the expected survival post-transplant using M2. For patients removed from the list (regardless of removal reason), *i.e.* those who died or were suspended during 2013, life years was the observed survival time from registration to removal date, death date or date of suspension. For a patient still on the waiting list on 1 January 2014, the life years was their observed survival time from registration to 31 December 2013 (including any period of suspension) plus their expected survival post-registration from the end of the simulation period using M1. Patient life years were then summed up across all 1,287 registrations to calculate the total population life years. The secondary outcomes included median time to transplant by aetiology, percentage of each aetiology transplanted, and impact on hepatocellular carcinoma (HCC) and variant syndrome (VS) cases.

### Monitoring the impact of the NLOS

Descriptive statistics were used. We present median (IQR) for continuous variables and n (%) for categorical variables. Comparisons between groups of interest were done using the $\chi^2$ or Fisher's exact tests for categorical variables, and the Kruskal-Wallis test for continuous variables. Survival from listing was estimated using Kaplan-Meier methods; survival time was the time from joining the transplant list to death, regardless of the length of time on the transplant list, whether or not the patient was transplanted, or any factors associated with such a transplant, *e.g.* donor type. Survival time was censored at either the date of removal from the list, or at the last known follow-up date post-transplant when no death date was recorded, or at the time of analysis if the patient was on the transplant list at the time of analysis. The log-rank test was used for between-group comparisons.

## Results

### The TBS as the optimal model to offer livers

The demographic and baseline characteristics of the M1 and M2 cohorts are presented in Table S2A,B while the status of patients at the end of follow-up is presented in Table S2C,D. The set of explanatory variables in M1 and M2 (cancer and non-cancer) is presented in Table 2, together with the C-statistics achieved by each model and 95% CIs. The C-statistics range between 0.62 and 0.73, indicating good predictive ability in all cases. Variables were selected based on statistical and clinical judgement.

Table 3 shows the two primary outcomes in the simulation. When comparing centre-based allocation – as operating in 2013 – with the three simulated scenarios, the number of patients predicted to die on the transplant list is reduced in both benefit- and need-based allocation, compared to those who actually died during 2013. Population life years were highest with benefit- and need-based allocation, and lowest with allocation as operating in 2013. These results led the working unit of the LAG to recommend the TBS as the optimal model to offer DBD livers to adult and large paediatric recipients with chronic liver disease (CLD) with UKELD >49 and those with HCC. It was also recommended to regularly update HCC and the clinical status of patients while waiting so that offering be done using as up-to-date data as possible. Consequently, since 2017, 'sequential data' on all patients registered to the elective liver transplant list have been submitted by transplant centres to NHSBT on a regular basis.

The impact of the offering models on patient aetiology is shown in Table S3, where the median waiting time to transplant for HCC is prolonged in M1 and M3 compared to the operating scheme in 2013, but is much shorter for patients with hepatitis C cirrhosis (HCV).

**Table 2. Recipient-, donor- and transplant-related explanatory variables selected (√) to model survival on the waiting list (M1) and survival post-transplantation (M2) for cancer (C) and non-cancer (NC) recipients.**

| Explanatory variable [units] (categories) | $M1_C$ | $M1_{NC}$ | $M2_C$ | $M2_{NC}$ |
|---|---|---|---|---|
| **Recipient-related** | | | | |
| Age [years] | √ | √ | √ | √ |
| Sex (male/female) | √ | √ | √ | √ |
| Presence of hepatitis C (yes/no) | √ | | √ | √ |
| Disease group[a] (2/…/10) | | √ | | √ |
| Serum creatinine [µmol/L] | √ | √ | √ | √ |
| Serum bilirubin [µmol/L] | √ | √ | √ | √ |
| INR | √ | √ | √ | √ |
| Serum sodium [mmol/L] | √ | √ | √ | √ |
| Potassium [mmol/L] | | | √ | √ |
| Albumin [g/L] | | | √ | √ |
| Renal support (yes/no) | √ | √ | √ | √ |
| Patient location (in-/out-patient) | √ | √ | √ | √ |
| Year of registration (2006/…/2012) | √ | √ | | |
| Previous abdominal surgery (yes/no) | | | √ | √ |
| Encephalopathy (yes/no) | | | √ | √ |
| Presence of ascites (yes/no) | | | √ | √ |
| Time on waiting list [days] | | | √ | √ |
| Diabetes (yes/no) | | | √ | √ |
| Maximum AFP level [IU/ml] | √ | | √ | |
| Maximum tumour size [cm] | √ | | √ | |
| Number of tumours (one/two/three+) | √ | | √ | |
| Interaction terms | | | | |
| Serum bilirubin * serum sodium | √ | √ | | |
| Disease group * serum bilirubin | | √ | | |
| Age * serum creatinine | | | √ | √ |
| Disease group * age | | | | √ |
| **Donor-related** | | | | |
| Age [years] | | | √ | √ |
| Cause of death (intracranial haemorrhage/road traffic accident/other trauma/other) | | | √ | √ |
| Body mass index [kg/m$^2$] | | | √ | √ |
| History of diabetes (yes/no/unknown) | | | √ | √ |
| Donor type (DBD/DCD)[b] | | | √ | √ |
| Interaction terms | | | | |
| Recipient presence of hepatitis C * donor history of diabetes | | | √ | √ |
| Recipient presence of hepatitis C * donor age | | | √ | √ |
| Donor type * recipient age | | | √ | √ |
| Donor type * recipient serum creatinine | | | √ | √ |
| Donor type * recipient disease group | | | | √ |
| **Transplant-related** | | | | |
| Donor-recipient blood group compatibility (identical/compatible) | | | √ | √ |
| Donor organ transplanted as whole (yes/no)[c] | | | √ | √ |
| **C-statistic (95% CI)** | 0.69 (0.64, 0.75) | 0.73 (0.71, 0.76) | 0.70 (0.64, 0.75) | 0.62 (0.59, 0.64) |

AFP, alpha-fetoprotein; INR, international normalised ratio.
The C-statistic was computed using Gönen and Heller's method.[11]
[a]According to Roberts *et al.*[14] liver disease categorisation, which maps up to three liver diseases provided as indications for transplantation at patient registration onto only one. For patient registrations between 1 January 2006 and 31 August 2007, disease group was determined from primary disease only as it was only possible to record one disease at registration during this time. The nine disease categories are (2) hepatitis C cirrhosis, (3) alcoholic liver disease, (4) hepatitis B cirrhosis, (5) primary sclerosing cholangitis, (6) primary biliary cirrhosis, (7) auto-immune or cryptogenic disease, (8) metabolic liver disease, (9) other, (10) re-transplant. Category (1) corresponds to the cancer group and is, therefore, not needed in NC models. Full details of the disease groups are presented in Table S4.
[b]Currently, this is a 'mute' characteristic as only donation after brain death livers are offered via the transplant benefit score.
[c]Donations after brain death from donors who are less than 40 years of age, weigh more than 50 kg and have stayed in an intensive therapy unit for less than 5 days meet the criteria for liver splitting.[12]

**Table 3. Mortality and patient-years associated with the operating allocation scheme in 2013 and the simulated allocation models based on the simulation cohort (1,287 registrations and 629 donors).**

| | N (%) died/removed[1] | Patient-years |
|---|---|---|
| Operating scheme in 2013 | 93 (7%) | 4,581 |
| Need (M1) | 48 (4%) | 5,187 |
| Utility (M2) | 95 (7%) | 4,779 |
| Transplant benefit (M3) | 48 (4%) | 5,262 |

[1]Removed due to clinical deterioration.

## The wider NLOS

Although not the focus of this paper, we briefly describe the processes that are currently followed in the UK to offer DBD livers to other patient groups *outside TBS offering*, and to offer DCD livers. Full details of patient registration and liver allocation criteria are documented in NHSBT online policies.[12,13]

Cases requiring emergency transplantation are registered using super-urgent transplantation criteria and offered DBD livers as top priority. For some recipients, the TBS does not adequately capture the need for transplantation; these individuals are classified as having VS. Examples of such indications are patients with poor quality of life associated with polycystic liver disease, severe pruritus or recurrent cholangitis when liver synthetic function is preserved, and certain patients with a genetic liver condition without associated CLD. Patients with VS are offered DBD livers in proportion to their annual representation of the total elective transplant registrant pool. At the time of writing, patients with VS are offered 10% of DBD livers following a probabilistic decision rule (if the liver is still available after offering to priority groups). A decision is made by the offering algorithm with 10% probability of offering to the pool of VS recipients, and 90% probability of offering to the pool of patients with CLD (including diuretic-resistant ascites and/or chronic hepatic encephalopathy groups) and HCC. The probability figure is regularly reviewed by NHSBT and endorsed by the LAG. If the donor liver is directed to the VS pool, patients are prioritised by waiting time on the list – the longest time first.

DCD offering did not change when NLOS was introduced to mitigate any unintended consequence of the implementation which might result in more prolonged cold ischaemic times and associated adverse outcomes for DCD grafts and recipients. A DCD organ is offered to the *local* centre first. If the local centre declines, the organ is offered to the other six centres in the country ordered by the centres' geographical proximity to the local centre and by the centres' recent levels of transplant activity.

Any within-fast-track-criteria donor liver (DBD or DCD) that is not accepted for a named patient or, when a DCD, not accepted by the local or neighbouring centre, is simultaneously offered to all centres via the fast-track offering scheme; a scheme designed to rapidly facilitate transplantation and foreshorten the offering process. Fast-track criteria (or *triggers*) are, in general terms, met when the liver is at risk of not being utilised within a reasonable time frame because of delays in the offering process or change of circumstances in patient acceptance. Triggers may also be met if there is a clear trend of repeated offer decline for a named patient because of concerns regarding organ quality. Fast-tracked livers may be accepted for any blood group-compatible or -identical registrant, meaning that fast-track offers are not made to named patients but to centres. It is a centre's decision to allocate the liver to a patient of their choosing (Figs S3 and S4 for DBD organs).

## Impact of the NLOS

We report on the performance of the NLOS in terms of deaths on the waiting list and total population life years (or surrogate metrics), as observed from 2 years' worth of data, and compare these outcomes with those in the 2 years preceding the implementation of the NLOS.

### *Death on the transplant list*

Table 4 shows the outcome, 6 months post-registration, for patients pre- and post-NLOS implementation. The mortality rate for patients on the list post-NLOS significantly fell compared with that of patients on the list pre-NLOS (4% and 8%, respectively; $\chi^2$ $p$ value <0.0001), supporting the NLOS development simulations.

### *Survival from listing as a surrogate to population life years*

The survival at 6 months post-registration for adult elective patients was statistically significantly greater for the post-NLOS period compared with pre-NLOS (log-rank $p$ value = 0.005; Fig. 1), in accord with the simulations run during NLOS development. The survival rates from listing for patients with CLD and HCC are shown in Fig. S5. Six-month survival post-listing is significantly greater post-NLOS compared with pre-NLOS for patients with CLD (log-rank $p$ value = 0.002), but this comparison is non-significant for patients with HCC. Interestingly, the survival rate at 6 months for patients registered post-NLOS was similar for the CLD and HCC groups whereas it was statistically significantly greater for patients with HCC compared to those with CLD in the pre-NLOS period (log-rank $p$ value = 0.0001).

**Table 4. Registration outcome for adult elective NHS Group 1 registrations on the UK liver transplant list.**

| Registration outcome at 6 months after joining the list | Pre-NLOS[1], n (%) | Post-NLOS[1], n (%) |
|---|---|---|
| Remained active/suspended | 394 (28) | 421 (28) |
| Died/removed due to clinical deterioration | 118 (8) | 66 (4) |
| Removed due to other reasons | 43 (3) | 34 (2) |
| Transplanted | 840 (60) | 959 (65) |
| Total | 1,395 (100) | 1,480 (100) |

NLOS, National Liver Offering Scheme.

[1]20 March 2016 to 19 September 2017 (pre) and 20 March 2018 to 19 September 2019 (post).

### *Registration activity*

A Fisher's exact test to examine the null hypothesis that urgency status (super-urgent or elective) and period (pre- or post-NLOS) are unrelated showed that urgency status did not differ over periods ($p$ >0.99; Table 5A). Equivalent investigations of aetiology of super-urgent registration, age group (elective adult or elective paediatric), graft number in elective adults (1$^{st}$, 2$^{nd}$,… graft), and type of patient (adult elective: CLD, HCC, HCC downstaging, VS,

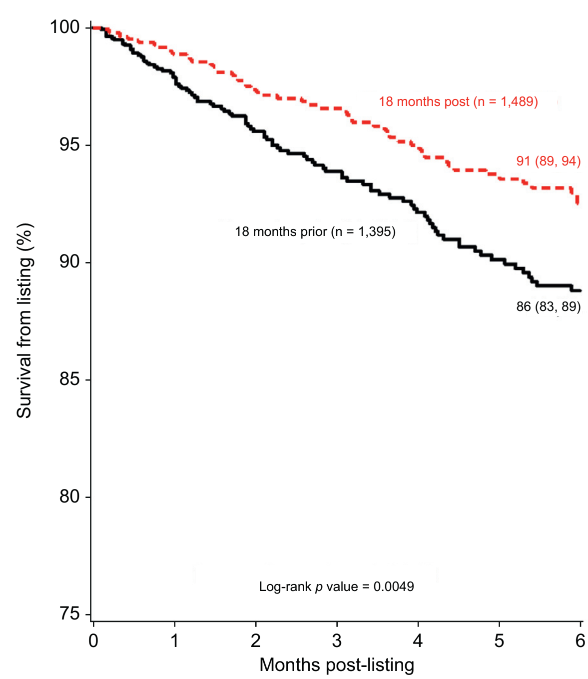


**Fig. 1. Six-month survival from listing for adult elective liver patients produced using Kaplan-Meier methods. To compare the survival between the pre-NLOS (18 months prior curve) and the post-NLOS (18 months post curve) groups, a log-rank test was conducted.** Patients were registered between 20 March 2016 and 19 September 2019; % survival (95% CI). (This figure appears in color on the web.)

**Table 5. NLOS monitoring metrics.**

| Monitoring metric | | Pre-NLOS | Post-NLOS |
|---|---|---|---|
| A. All NHS Group 1 liver registrations in the UK (n = 2,314 pre-NLOS; n = 2,411 post-NLOS) | No. elective (%) | 2,053 (89) | 2,139 (89) |
| | No. super-urgent (%) | 261 (11) | 272 (11) |
| B. DBD donors (n = 1,970 pre-NLOS; n = 2,144 post-NLOS) | No. livers offered for donation (%) | 1,799 (91) | 2,003 (93) |
| C. DCD donors (n = 2,320 pre-NLOS; n = 2,498 post-NLOS) | No. livers offered for donation (%) | 1,900 (82) | 2,034 (81) |
| D. All DBD liver-only offers (all offers; some ultimately not transplanted) | No. offers (no. donors) | 5,487 (1,799) | 8,918 (2,003) |
| | Median number (IQR) of offers per donor | 2 (1, 5) | 3 (1, 7) |
| E. DBD liver-only offers (all offers) that were ultimately transplanted (n = 3,256 pre-NLOS; n = 5,085 post-NLOS) | No. declined offers (%) | 1,614 (50) | 3,037 (60) |
| | No. accepted but not used offers (%) | 139 (4) | 500 (10) |
| | No. transplanted offers (%) | 1,503 (46) | 1,548 (30) |
| F. Adult elective liver and liver/kidney DBD donor transplants[1] (n = 1,543 total transplants pre-NLOS; n = 1,586 post-NLOS) | No. (%) | 1,201 (78) | 1,232 (78) |
| | Median (IQR) time to transplant [days] | 77 (26, 217) | 40 (10, 140) |
| | Median CIT [hours] | 8.6 | 9.0 |
| G. Adult elective liver and liver/kidney DCD donor transplants[2] (n = 408 total transplants pre-NLOS; 366 post-NLOS) | No. (%) | 393 (96) | 357 (98) |
| | Median (IQR) time to transplant [days] | 61 (24, 174) | 52 (20, 130) |
| | Median CIT [hours] | 7.5 | 7.4 |

Offering and transplanting of deceased donor livers in the pre-NLOS (20 March 2016 to 19 March 2018) and post-NLOS (20 March 2018 to 19 March 2020) periods.
CIT, cold ischaemic time; NLOS, National Liver Offering Scheme.
[1]The majority of these transplants were liver-only (1,172/1,201 pre-NLOS; 1,205/1,232 post-NLOS).
[2]The majority of these transplants were liver-only (392/393 pre-NLOS; 356/357 post-NLOS).

hepatoblastoma, liver/cardiothoracic; paediatric elective: hepatoblastoma, non-hepatoblasma, liver/cardiothoracic) showed no significant association between the pre- and post-NLOS periods and these categorisations (data not shown), indicating lack of evidence that registration of new patients statistically significantly changed pre- and post-NLOS.

The monitoring results of *Offering* and *Transplant Activity* are presented in the supplementary data.

## Discussion

This paper describes the development and implementation of the NLOS. Since its inception, the NLOS has seen nearly 70% of DBD livers prioritised for offering via the TBS (Fig. S6), in line with how the NLOS was designed. To our knowledge, it is the first scheme that offers organs based on statistical prediction of transplant benefit. Schemes used in other countries differ from the TBS, mostly offering purely on need for transplantation, and make direct comparisons difficult. In the US, the key metric for liver allocation is the MELD score, where certain patient populations, such as those with HCC, may be granted MELD exception points.[15,16] Most European countries operate the MELD scoring system, too, the exception being Scandiatransplant who allocate based on clinical waiting time.[17]

The authors recognise the limitations of the work. The gestation of the TBS was long. It was then implemented as part of a large programme that transformed transplant processes and systems in NHSBT over a period of 5 years. The TBS was introduced as part of a large overhaul of liver transplant processes in March 2018 which saw many elements impacted; from data collection forms to national policies and reporting paradigms. It was agreed, prior to its introduction, that the NLOS would be monitored and updated on a regular basis – with parameter estimates updated as more data accrue – to ensure it accurately reflects up-to-date practice and to provide the opportunity to mitigate against any unintended consequence of the scheme. Liver offering based on patient benefit represents progress relative to the former needs-based offering but it does not come without challenges. There is perception that certain patient groups are disadvantaged and that the benefit to the *individual* can be at odds with the good of the waiting list *population*.[18] Also, the TBS does not fit all patients, such as VS. The NLOS monitoring exercises led the LAG to recommend, in 2021, an update of the parameter estimates in the TBS. A working group was formed. The cohort for the new analyses included data up to 2016, updating the intelligence provided by the UKTR. The recommendations (outside of the scope of this paper) by the group included the use of grouped registration year instead of individual registration years in M1, and more stringent model inclusion criteria of explanatory variables in both M1 and M2. The primary directive for the optimal approach was, once again, to minimise the number of deaths on the list and to maximise patient life years gained but balanced to serve possibly disadvantaged patients such as those with HCC, where concerns had been raised that prediction of transplant free survival (M1) was overestimated within the models.[19] The latest version of the TBS was introduced in the UK in October 2022 and continues to be refined in light of monitored performance.

The implementation of the scheme has generated considerable debate amongst stakeholders including patient groups and transplant clinicians. Concerns have been expressed about the restriction in recipient selection when offers are made and the perceived disadvantage of patients who have been less prioritised within the scheme, which is associated with increased waiting times. This was never an intended objective of the scheme, where survival on the list is just as valuable as survival post-transplantation but highlights the challenge of ensuring that communities both understand the principles that underpin this complex offering system and have realistic expectations on whether it delivers on a *population* as well as *individual* basis.

The novelty of the NLOS relates to the utility component (M2) built into the offering process. Predicted utility in the scheme is estimated as 5-year patient survival after transplantation. This length of follow-up was selected as data were readily available from the UKTR for a current development cohort, while longer follow-up was not. There was also uncertainty regarding accurately predicting outcomes beyond 5 years where mortality is decreasingly directly associated with the transplant. Whether this is the optimal measure of utility has not been tested and will need to be considered as the scheme is taken forward. Similarly, the potential incorporation of measures of quality of life, which matter to patients and provide a distinct perspective on the benefits of transplantation,[18] will need to be discussed.

Other approaches to the development of offering schemes have been explored in the literature, such as estimating the potential benefit outcomes whilst accounting for organ scarcity, and employing machine learning to improve donor-recipient matching.[20,21] A recent review of the ethics of donor organ allocation[25] discussed the tension between prioritisation by need and benefit, and highlighted the importance of transparency in the chosen process.

In conclusion, we have demonstrated the multiple steps followed in establishing the principles of a new liver offering scheme, including engaging with stakeholders and following a statistical modelling approach based on the well-curated UKTR database. Such a scheme currently offers DBD organs nationally on the basis of transplant benefit. We provide preliminary evidence to support that the scheme has delivered on its objective of reducing waiting list mortality while preserving post-transplant outcomes and, in so doing, maximising life years gained from liver transplantation for the waiting list *population*.


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## Abbreviations

C, cancer; CLD, chronic liver disease; DBD, donation after brain death; DCD, donation after circulatory death; HCC, hepatocellular carcinoma; HCV, hepatitis C cirrhosis; IPCW, inverse probability of censoring weighted; LAG, liver advisory group; MI, multiple imputation; NC, non-cancer; NHSBT, NHS Blood and Transplant; NLOS, National Liver Offering Scheme; TBS, transplant benefit score; UKELD, UK end-stage liver disease; UKTR, UK Transplant Registry; VS, variant syndrome.


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## Conflict of interest

The authors of this study declare that they do not have any conflict of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

## Authors' contributions

EA, RT, DT and AG were part of the working groups that conceived, developed and introduced the National Liver Offering Scheme (NLOS) in the UK. EA and RT carried out the statistical analyses and wrote the manuscript. DT and AG planned and edited the manuscript.

## Data availability statement

The data that have been used were submitted by NHSBT and hospital staff (including Specialist Nurses for Organ Donation, National Organ Retrieval Service personnel, transplant surgeons, recipient transplant coordinators and other hospital staff) to the UK Transplant Registry (UKTR). Information on how to access data from the UKTR can be found online at https://www.odt.nhs.uk/statistics-and-reports/access-data/ [Accessed: 2 February 2024].


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## Supplementary data

Supplementary data to this article can be found online at http://doi:10.1016/j.jhep.2024.03.020.

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